

PRATIBANDH: SECURE DIGITAL DOCUMENT & EVIDENCE VAULT

Comprehensive System Architecture, Technical Specifications & Component Directory

STATUTORY EVIDENCE VAULT | Section 65B Indian Evidence Act & IT Act 2000 Compliant

Problem Statement: SIH 1690 (Centralized Encrypted Forensic Repository) | Build Version: 2.4.0 (Gov-Release)

1. EXECUTIVE SUMMARY & CORE OBJECTIVES

PRATIBANDH is a government-grade digital evidence and case document repository built for police departments, courts, and forensic laboratories. It solves the critical operational risks of document tampering, unauthorized screen scraping, untracked access, and inter-jurisdictional data silos through four core pillars:

- 1. Dynamic Spotlight Masking:** Documents are rendered onto an HTML5 Canvas behind a 10px Gaussian blur. Only a 130px circular aperture following the cursor is rendered sharp in real time, defeating full-page photographic capture.
- 2. Forensic Pixel Watermarking:** Officer identity (Name, Form/Badge ID, IP address, and ISO timestamp) is repeatedly baked directly into canvas pixels at -25°, making any smartphone photo capture traceable to the exact terminal.
- 3. AI-Powered ID Card & Facial Biometrics:** Client-side WebAssembly OCR (Tesseract.js) and 128-dimensional facial biometric embedding neural nets (@vladmandic/face-api) for tamper-proof officer authentication.
- 4. AES-256-CBC Cryptography & Statutory Audit Ledger:** Zero plaintext files at rest; immutable record-by-record ledger compliant with Section 65B of the Indian Evidence Act.

2. HIGH-LEVEL ARCHITECTURE & DATA FLOW

[CLIENT TIER : React 18 + Vite SPA] ■■■ Presentation: GovHeader (IST clock), GovFooter, FloatingActions Hub, RBAC Sidebar ■■■ Security Engine: Dual-Canvas HTML5 Spotlight Renderer + Pixel Watermark Stamping ■■■ AI Computer Vision: Tesseract.js OCR (Wasm) + vladmandic/face-api (128-d vectors) ■ HTTPS REST (Axios) / WSS (Socket.IO) ▼ [APPLICATION SERVER TIER : Node.js + Express.js] ■■■ Middleware Pipeline: Helmet (CSP/HSTS) | CORS | Morgan | Multer Multipart ■■■ Auth & RBAC: JWT Bearer Verification + roleCheck (super_admin, station_admin, officer) ■■■ Controllers: Auth, RegisteredIDs, Cases, Documents, Collaboration, Audit, Dashboard ■■■ Audit Interceptor: Immutable capture of action, resource, IP, user-agent, timestamp ■■■ [DATABASE : MongoDB Atlas] [CRYPTOGRAPHIC STORAGE : Disk] - Users & RegisteredIDs - AES-256-CBC Encrypted Files (*.pdf.enc) - Cases & Evidence Records - 16-byte Random IV Buffer Header - AuditLogs (Indexed Ledger) - Ephemeral Decryption Buffer Pipeline

3. BIT-BY-BIT TECHNOLOGY STACK & JUSTIFICATION

Layer / Area	Technology	Role & Functionality	Why It Was Chosen
Frontend UI	React 18 + Vite 5	Reactive SPA component architecture, fast HMR, state-driven rendering.	Virtual DOM allows smooth 60fps canvas cursor tracking without lag.
Styling & Theme	TailwindCSS 3.4	Custom Indian Gov navy (#0a1b38) & gold (#d4af37) palette, responsive grid.	Utility classes enforce strict aesthetic consistency across all views.
AI OCR Engine	Tesseract.js 5.1 (Wasm)	Client-side character recognition from camera frames & photos.	Zero server GPU load; extracts Regd/Form Nos directly inside user browser.
Facial Biometrics	@vladmandic/face-api	TinyFaceDetector + 68 landmarks + 128-d descriptor extraction.	Privacy-first biometric extraction; video streams never leave client.
Document Viewer	HTML5 2D Canvas API	Dual-canvas pipeline for spotlight masking & forensic pixel watermarks.	Prevents DOM text inspection; blocks screenshot scraping completely.
Backend API	Node.js + Express 4.18	Asynchronous REST API routing, streaming endpoints, middleware pipeline.	High-throughput non-blocking I/O for concurrent evidence downloads.

Database & ODM	MongoDB + Mongoose	Document storage with text indexes, virtual aliases, and aggregation pipes.	Flexible schemas for complex crime metadata and sub-millisecond query speed.
File Encryption	Node Crypto (AES-256)	AES-256-CBC cipher with unique 16-byte IV prepended to every file.	Military-grade file security at rest; raw bytes unreadable if drive is stolen.
Auth Security	BcryptJS + JWT 9.0	Password hashing (12 rounds) & 24h HMAC-SHA256 signed bearer tokens.	Protects passwords from rainbow table attacks; enables stateless RBAC.
Real-time Gateway	Socket.IO 4.7	WebSocket event bus with station-specific room isolation.	Instantly pushes case collaboration and sharing alerts between stations.
HTTP Security	Helmet 7.1 + CORS	Applies CSP, HSTS, X-Frame-Options, and cross-origin security rules.	Mitigates Clickjacking, Cross-Site Scripting (XSS), and MIME sniffing.

4. BIT-BY-BIT MODULE EXPLANATION & FUNCTIONALITY

4.1 Anti-Screenshot Spotlight Document Viewer (DocumentViewer.jsx)

- **Dual-Canvas Pipeline:** The component instantiates an offscreen in-memory canvas where sharp document text, classification stamps, and case seals are drawn.
 - **Spotlight Masking:** The primary visible canvas draws the offscreen canvas with a 10px Gaussian blur (ctx.filter = 'blur(10px)'). When hovering, ctx.clip() cuts a 130px circular aperture at cursor position (X, Y) rendering the sharp version.
 - **Forensic Pixel Stamping:** Rotated (-25°) watermark lines containing CONFIDENTIAL • OFFICER NAME • FORM NO • IST TIMESTAMP are repeatedly stamped directly into canvas pixels.
 - **Hardware Capture Interception:** Keydown listeners block Ctrl+P, Ctrl+S, Ctrl+U, PrintScreen, and the right-click context menu.
- Why Used: Standard HTML text viewers allow full-page screenshots and DOM text scraping. Canvas spotlighting restricts visibility to an un-scrappable 130px aperture while stamping traceable forensic credentials into every pixel.*

4.2 AI ID Card Verification & Biometric Login (IDVerification.jsx, ocr.js, faceApi.js)

- **Image Preprocessing:** Upscales camera frames to ≥ 1200px, computes luminosity grayscale ($Y = 0.299R + 0.587G + 0.114B$), and applies +30% contrast stretching for maximum OCR character clarity.
 - **OCR & Barcode Extraction:** Tesseract.js (Wasm) and native BarcodeDetector API extract Form Number and personnel name tokens.
 - **128-d Biometric Descriptors:** TinyFaceDetector locates the face on the ID card; FaceRecognitionNet computes a 128-dimensional floating point vector embedding.
 - **Bio-Data Matcher:** Backend route /api/auth/match-id-card matches extracted registration numbers and tokens against the database, calculating confidence scores (≥ 95%).
- Why Used: Eliminates password sharing and stolen credential attacks by requiring visual, verifiable physical departmental ID verification.*

4.3 Statutory Audit Trail & Activity Analytics (AuditLog.jsx, server/routes/audit.js)

- **Visual Activity Bar Chart:** Ranks officers by 'Times Opened', 'Doc Views', 'Downloads', 'Logins', and 'Modifications' with multi-colored segmented proportional bars.
 - **Tamper-Evident Ledger:** Tabular historical audit list showing IST timestamps, officer identity, Form Number, action badge, target resource, details, and IP address.
 - **CSV & Print Export:** One-click generation and download of official CSV audit logs compliant with Section 65B of the Indian Evidence Act.
 - **Database Aggregation:** Endpoint /api/audit/user-stats utilizes MongoDB \$lookup, \$unwind, and \$group pipelines to aggregate activity counts directly on the database engine.
- Why Used: Guarantees digital evidence admissibility in court by proving an unbroken, certified digital chain of custody.*

4.4 AES-256-CBC File Encryption Pipeline (server/utills/encryption.js)

- **Encryption (encryptFile):** Reads uploaded file buffer, generates a random 16-byte IV via crypto.randomBytes(16), encrypts payload using aes-256-cbc, and prepends IV to the ciphertext buffer.

- **Decryption (decryptFile):** Slices first 16 bytes (IV), decrypts payload to a temporary file, streams to authorized officer, and immediately deletes temporary file via fs.unlinkSync.

Why Used: Ensures that if server hard drives or backups are physically stolen, zero unencrypted evidence is compromised.

4.5 Case Management & Collaboration (CaseList.jsx, SharedCases.jsx)

- **Case Dossier Management:** Auto-generates unique IDs (CASE-YYYY-XXXXX), tracks crime categories (Murder, Rape, Theft, Cybercrime, Fraud, etc.), suspects, victims, priority, and assigned officers.
- **Cross-Station Sharing:** Dispatches inter-station access requests (/api/collaboration), enabling multi-jurisdictional collaboration across police stations.

Why Used: Eliminates jurisdictional silos during multi-state criminal investigations.

4.6 Government Frame & Floating Action Hub (GovHeader.jsx, FloatingActions.jsx)

- **GovHeader & GovFooter:** Displays National Tricolor Ribbon, Ashoka Lion Emblem, Ministry of Home Affairs branding, live Indian Standard Time (IST) clock, and 1930 Cyber Crime Helpline.
- **FloatingActions:** Speed-dial menu offering one-click navigation to GitHub, Audit Charts, AI Scanner, Evidence Upload, and Case Creation.

Why Used: Enforces statutory credibility and maximizes operational speed during critical incident handling.

5. DATABASE SCHEMAS & STORAGE ARCHITECTURE

```
• User Schema (User.js): - name, email (unique), password (bcrypt hash, 12 rounds), role (super_admin, station_admin, officer, etc.) - station, department, formNumber/badgeId (unique, indexed), phone, isActive, lastLogin
• Case Schema (Case.js): - caseId (unique: CASE-YYYY-XXXXX), title, description, crimeType (enum), status, priority - suspects [{name, age, gender, description}], victims [{name, age, gender}] - assignedOfficers [ObjectId -> User], station, courtName, judge, sharedWith [{station, accessLevel}] - createdBy [ObjectId -> User], timestamps (createdAt, updatedAt)
• Document Schema (Document.js): - docId (unique: DOC-YYYY-XXXXX), title, docType (fir, charge_sheet, evidence, forensic_report, etc.) - case [ObjectId -> Case], filePath (encrypted .enc path), originalName, mimeType, fileSize - isEncrypted (Boolean), uploadedBy [ObjectId -> User], tags [String], ocrText, accessRestriction
• AuditLog Schema (AuditLog.js): - user [ObjectId -> User], action (view, download, upload, edit, delete, login, share) - resourceType (document, case, user, system), resourceId, details, ipAddress, userAgent, timestamp
• RegisteredID Schema (RegisteredID.js): - formNumber (unique, indexed), name, email, role, station, department, phone - idCardImage (.enc path), faceDescriptor ([128 Numbers]), addedBy [ObjectId -> User], isActive
```

6. STATUTORY & REGULATORY COMPLIANCE

- **1. Section 65B, Indian Evidence Act (1872):** Produces certified computer-generated evidence audit trails recording originating IP, officer credentials, and exact microsecond timestamps required for court admissibility.
- **2. Information Technology Act (2000):** Encrypted storage and RBAC controls comply with Section 43A (compensation for failure to protect sensitive data) and Section 69A.
- **3. ISO/IEC 27001 Information Security:** Implements strict role-based access, cryptographic key management, and zero-plaintext storage architecture.
- **4. CCTNS Interoperability:** Standardized crime classification schemas allow direct integration with national Crime and Criminal Tracking Network & Systems.